Inflation Persistence After an Energy-Price Shock: New Keynesian Model with Vertical Production Networks and Real Wage Rigidity*

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Abstract

Abstract here.

Keywords: New Keynesian model, Production network, real wage rigidity

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*COMMENTS. All remaining errors are my own.

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1 Introduction

The energy crisis in 2022 led to a surge in natural gas prices, which has in turn caused a significant increase in inflation in the UK. Whereas the natural gas price shock was transitory, the inflation response was lagged and persistent: the peak of the inflation response occurred a few months later than that of the natural gas price, and the inflation rate remained elevated for a long time after the natural gas price returned to normal. TODO: present observations!

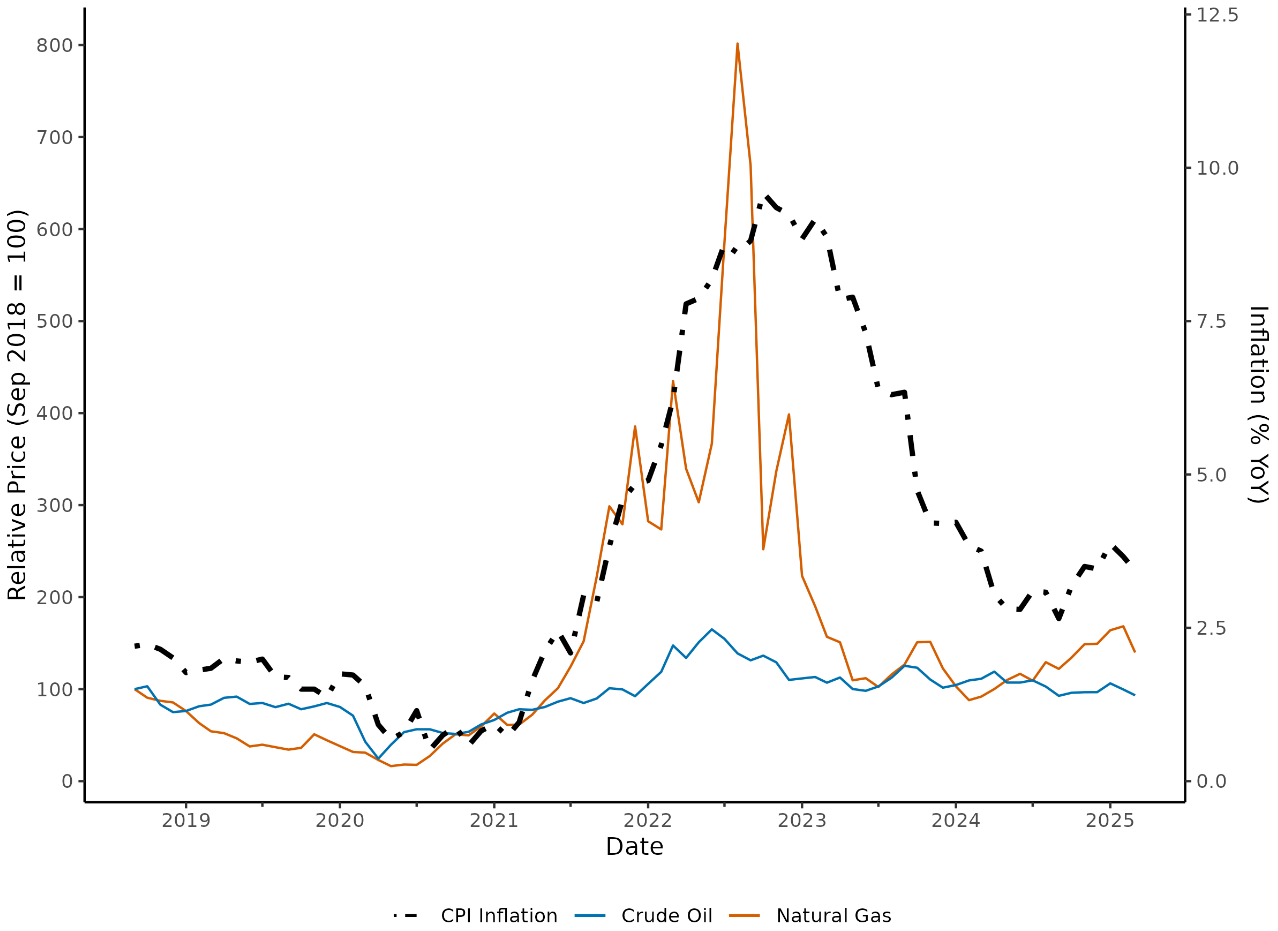

Figure 1: Energy Prices and UK CPI Inflation between September 2018 and March 2025

Notes: Solid lines show GBP-denominated natural gas (EU) and Brent crude oil prices indexed to September 2018 = 100 (left axis). Black dashed line shows headline CPI inflation, year-on-year (right axis). Sources: FRED

The standard New Keynesian model for a small open economy which imports energy as a factor of production at an exogenous price (hereafter "the benchmark model"; see Appendix C for details) fails to replicate this hump-shaped and persistent response of inflation to the energy price shock.

I extend the benchmark model in the three following ways to replicate a hump-shaped and persistent inflation response to the energy price shock. Intuitively WHY? Describe the sense of economics First, I introduce production networks with an upstream industry that utilises imported energy and labour to produce intermediate goods and a downstream industry using the intermediate goods and labour to produce final goods that are consumed by households. Firms in both upstream and downstream industries adopt Calvo pricing for their produced goods with different Calvo probability parameters between the two industries. There is a single labour market for both industries, where workers move freely between the two industries. Consequently, the wage is identical across the two industries in the equilibrium.

Second, in this production network, the downstream industry is myopic in its pricing decisions (see Minton and Wheaton (2023)). While non-myopic Calvo-pricing firms incorporate full information about the expected future marginal cost into their reset price calculations, myopic firms discount the future expected marginal cost and hence put more weight on the current marginal cost when setting their prices. The myopia of the downstream industry corresponds to the difficulty, particularly for downstream firms, to perfectly foresee the future path of energy price, compared to the upstream industry. TODO: more details here.

Third, I introduce real wage rigidity in the labour market, which represents the friction in the labour market arising from fixed contracts, employment regulations, and recruitment costs.

The rest of the paper is organised as follows. ORGANISE! The related literature is reviewed in the following subsection. In Section 2, I describe the benchmark model and its extensions, as well as calibrate the model parameters for the simulation. In Section 3, I present the simulation results and evaluate them by conducting sensitivity analysis. Finally, the paper concludes in Section 4.

1.1 Literature

Gagliardone and Gertler (2025) explores AAA.

Gomellini et al. (2025) explores BBB.

- Production networks
- Real wage rigidity
- Oil shock `https://www.nber.org/papers/w13368`

2 The Model

This section describes the paper's main model. Although the model is extended upon the "benchmark model", the section is self-contained. The details of the "benchmark model" is presented in Appendix C.

2.1 Economic Environment and Terms-of-Trade Shock

Consider a small open economy that imports a quantity Z_t of energy from the rest of the world. Assume the economy does not produce energy domestically, and its domestic demand for energy does not affect the global energy price. There is also a homogeneous internationally tradable good that the economy can produce and export. Assume without loss of generality that the domestic economy can produce E_t units of this traded good using E_t units of domestic labour, and the market for the traded good is perfectly competitive. This implies that the domestic-currency price of the traded good is equal to the nominal wage W_t, the nominal marginal cost of production. Define the unit price of energy imports in domestic currency to be S_t, and denote the relative price of imports to exports (i.e., the terms of trade) as $s_t = S_t/W_t$. The small open economy takes this terms of trade s_t as given, and shocks to s_t represent energy price shocks in the analysis.

Assume that the terms of trade s_t follows an exogenous AR(1) process. Let $\hat{s}_t$ denote the log-deviation of s_t from its steady state value $\bar{s}$ which is normalised to one without loss of generality, so $\hat{s}_t$ follows

$$\hat{s}_t = \rho \hat{s}_{t-1} + u_t, \tag{1}$$

where u_t is a zero-mean i.i.d. shock to the terms of trade, and ρ measures the persistence of the exogenous shock.

All goods are perishable, and the model does not have financial assets to be traded internationally, so net exports must be zero in the balance-of-payments equilibrium. Mathematically, this means $W_t E_t = S_t Z_t$, or equivalently,

$$E_t = s_t Z_t. \tag{2}$$

2.2 Production Network

Apart from the production of tradable goods (which utilises E_t units of labour to produce the same quantity of the tradable goods), the economy entails a production network of domestic consumption goods, which consists of upstream and downstream industries that are vertically structured. Both industries use domestic labour as an input, which does not require any industry-specific skills, and the workers are completely mobile between the two industries, so that the wage is identical across the two industries in the equilibrium.

In the upstream industry, there is a continuum of firms $j \in [0, 1]$ that uses imported energy and domestic labour to produce differentiated intermediate goods, which will be used as inputs by the downstream industry. The production of the upstream industry firm j is described by the following Leontief production function:

$$Q_t(j) = \min \left\{ \frac{Z_t(j)}{\alpha^U}, \frac{N_t^U(j)}{1 - \alpha^U} \right\}, \tag{3}$$

where $Q_t(j)$ denotes the output of firm j in the upstream industry, $Z_t(j)$ and $N_t^U(j)$ denote the usage of energy and labour by firm j as inputs, and the parameter $\alpha^U \in (0, 1)$ is the share of energy in the production function.[1] The production function is in Leontief form, which implies that there is no substitutability between energy and labour in producing the intermediate goods. The production cost is composed of the cost of labour and energy, whose

[1] I use a superscript U (D) to indicate that a variable is about the upstream (downstream) industry.

nominal prices are W_t and S_t, respectively. As the upstream industry firm minimises its production cost, energy and labour are used in fixed proportions:

$$Z_t(j) = \alpha^U Q_t(j), \quad N_t^U(j) = (1-\alpha^U)Q_t(j). \tag{4}$$

The nominal marginal cost X_t^U of producing one unit of intermediate good is constant across firms in the upstream industry and is given by

$$X_t^U = \alpha^U S_t + (1-\alpha^U)W_t = (1-\alpha^U + \alpha^U s_t)W_t. \tag{5}$$

It follows that the real marginal cost $x_t^U = X_t^U / P_t^U$ is

$$x_t^U = \left(1 - \alpha^U + \alpha^U s_t\right) \cdot \frac{W_t}{P_t^D} \cdot \frac{P_t^D}{P_t^U} = (1 - \alpha^U + \alpha^U s_t) \cdot w_t \cdot \frac{P_t^D}{P_t^U}, \tag{6}$$

where $w_t = W_t / P_t^D$ is the real wage.[2] Since $\bar{s} = 1$ in steady state, the parameter α^U can be interpreted as the steady-state energy share of the upstream industry firm's production costs. The basket of differentiated intermediate goods is aggregated into a composite intermediate good using a CES aggregator:

$$Q_t = \left(\int_0^1 Q_t(j)^{\frac{\varepsilon-1}{\varepsilon}} dj\right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{7}$$

which is supplied to the downstream industry to be used in the production of consumption goods. Here, ε is the elasticity of substitution within the upstream industry intermediate goods.[3]

In the downstream industry, there is a continuum of firms $i \in [0,1]$ that uses the composite intermediate goods produced by the upstream industry as well as domestic labour to produce differentiated final goods, which are consumed by households. The production of the downstream industry firm i is also in Leontief form:

$$Y_t(i) = \min\left\{\frac{M_t(i)}{\alpha^D}, \frac{N_t^D(i)}{1-\alpha^D}\right\}, \tag{8}$$

where $Y_t(i)$ denotes the output of firm i in the downstream industry, $M_t(i)$ and $N_t^D(i)$ denote the usage of intermediate goods composite and labour by firm i as inputs, and the parameter $\alpha^D \in (0,1)$ is the share of intermediate goods composite in the production function.[4] Sim-

[2]While the real term of the upstream marginal cost is measured by the price of intermediate goods P_t^U, I measure wage, which is identical across both industries in equilibrium, by the price of final consumption goods P_t^D.

[3]For simplicity, the model assumes the elasticity of substitution is the same for both upstream and downstream industries.

[4]α^D can be interpreted as the (D, U)-element of the input-output matrix in a more generalised production network model.

ilarly to the production of the upstream industry, the production cost minimisation implies that intermediate goods and labour are used in fixed proportions:

$$M_t(i) = \alpha^D Y_t(i), \quad N_t^D(i) = (1-\alpha^D) Y_t(i). \tag{9}$$

The nominal marginal cost X_t^D of producing one unit of the final consumption good is

$$X_t^D = \alpha^D P_t^U + (1-\alpha^D) W_t, \tag{10}$$

and the real marginal cost $x_t^D = X_t^D / P_t^D$ is

$$x_t^D = \left(1 - \alpha^D + \alpha^D \frac{P_t^U}{W_t}\right) \frac{W_t}{P_t^D} = \left(1 - \alpha^D + \alpha^D \frac{P_t^U}{W_t}\right) w_t. \tag{11}$$

Finally, domestic households consume the CES aggregated basket of differentiated final goods:

$$C_t = \left(\int_0^1 C_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{12}$$

where $C_t(i) = Y_t(i)$ holds in the equilibrium for all i as the market clears. Solving the household's utility maximisation problem under the budget constraint yields the demand for each differentiated final good i:

$$Y_t(i) = \left(\frac{P_t^D(i)}{P_t^D}\right)^{-\varepsilon} Y_t, \tag{13}$$

where

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}}, \quad P_t^D = \left(\int_0^1 P_t^D(i)^{1-\varepsilon} di\right)^{\frac{1}{1-\varepsilon}}. \tag{14}$$

Similarly, solutions to the downstream industry's cost minimisation problem yield the demand for each intermediate good j by the downstream industry:

$$M_t(j) = \left(\frac{P_t^U(j)}{P_t^U}\right)^{-\varepsilon} M_t, \tag{15}$$

where $M_t = \int_0^1 M_t(i) di$ is the downstream industry's aggregate demand for the intermediate goods basket. In the equilibrium, the market for intermediate goods clears for each differentiated intermediate good j, so $M_t(j) = Q_t(j)$ for all j.

Since net exports are zero and there is no investment or government expenditure in the model, the real GDP is measured by the aggregate output of the downstream industry Y_t, which is equal to aggregate consumption C_t in the equilibrium. Since consumers only consume final goods and do not directly consume intermediate goods or energy, CPI inflation is determined solely by the price level of final goods:[5]

$$\pi_t = \pi_t^D = \frac{P_t^D - P_{t-1}^D}{P_{t-1}^D}. \tag{16}$$

2.3 Market Clearing and Aggregation

Let Z_t denote the aggregate demand for energy by the downstream industry and N_t^D and N_t^U denote the aggregate demand for labour by the downstream and upstream industry, respectively:

$$Z_t = \int_0^1 Z_t(j)dj, \quad N_t^D = \int_0^1 N_t^D(i)di, \quad N_t^U = \int_0^1 N_t^U(j)dj. \tag{17}$$

Then, the total demand for labour in the economy L_t is the sum of the labour demand by the two industries as well as the competitive firms producing the tradable goods:

$$L_t = N_t^D + N_t^U + E_t. \tag{18}$$

By substituting the demand functions for intermediate ((15)) and final goods ((13)), one can obtain the aggregate demands for the intermediate goods M_t and labour N_t^D for the downstream industry and energy Z_t and labour N_t^U for the upstream industry:

$$M_t = \alpha^D \Delta_t^D Y_t, \quad N_t^D = (1-\alpha^D)\Delta_t^D Y_t, \quad Z_t = \alpha^U \Delta_t^U Q_t, \quad N_t^U = (1-\alpha^U)\Delta_t^U Q_t, \tag{19}$$

where $\Delta_t^D = \int_0^1 \left(P_t^D(i)/P_t^D\right)^{-\varepsilon} di$ and $\Delta_t^U = \int_0^1 \left(P_t^U(j)/P_t^U\right)^{-\varepsilon} dj$ are the usual price dispersion terms for the downstream and upstream industry, respectively. Substituting the balance of payments condition (2) into the labour demand equation (18) gives $L_t = N_t^D + N_t^U + s_t Z_t$. Further substituting the aggregate demands derived above yields:

$$L_t = N_t^D + N_t^U + s_t Z_t \tag{20}$$

$$= [1 - \alpha^D + (1 - \alpha^U + \alpha^U s_t)\alpha^D \Delta_t^U]\Delta_t^D Y_t, \tag{21}$$

which can be rearranged to express the aggregate output Y_t as a function of the total labour demand L_t and the terms of trade s_t:

$$Y_t = \frac{L_t}{[1 - \alpha^D + (1 - \alpha^U + \alpha^U s_t)\alpha^D \Delta_t^U]\Delta_t^D}. \tag{22}$$

This implies that the economy's effective labour productivity is

$$\frac{Y_t}{L_t} = \frac{1}{[1 - \alpha^D + (1 - \alpha^U + \alpha^U s_t)\alpha^D \Delta_t^U]\Delta_t^D}, \tag{23}$$

[5] For this reason, I hereafter drop the superscript D to emphasise the dynamics of the entire economy, while keep it for describing the downstream industry specifically, but they are the same.

which declines as either or both of the industry-wise relative-price distortions Δ_t^D and Δ_t^U increase, and as the terms of trade s_t depreciates (i.e., imported energy price increases in domestic currency). Mathematically, the relative-price distortion in final goods Δ_t^D has a larger negative effect on the economy's effective labour productivity than that in intermediate goods Δ_t^U, which is intuitive because the former directly affects the price of final goods, while the latter only indirectly affects it through the cost of intermediate goods. When the terms of trade s_t depreciates, this means that the relative price of energy imports increases compared to other traded goods, which will require more labour input to produce the tradable goods E_t to maintain the balance of payments equilibrium. Consequently, given the fixed amount of total labour in the economy L_t, less labour can be allocated to the production of intermediate and final consumption goods, which is equivalent to a decline in the economy's effective labour productivity.

2.4 Households and Labour Supply

As per the standard New Keynesian framework, households with CRRA utility from consumption and disutility from labour supply maximise the expected discounted sum of utility subject to the budget constraint:

$$\max_{\{C_t,B_t,L_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-1/\sigma}}{1-1/\sigma} - \frac{L_t^{1+1/\psi}}{1+1/\psi} \right] \tag{24}$$

$$\text{s.t. } P_t^D C_t + B_t = (1+i_{t-1})B_{t-1} + W_t L_t + \Pi_t, \tag{25}$$

where β, σ, and ψ are the parameters of discount factor, elasticity of intertemporal substitution, and Frisch elasticity of labour supply, respectively, while B_t is the holding of nominal one-period bond, i_t is the nominal interest rate, and Π_t is the nominal profit from owning firms. FOCs with respect to C_t and L_t yield the Euler equation and the labour supply equation, respectively:

$$\frac{C_t^{-1/\sigma}}{P_t^D} = \beta(1+i_t)\mathbb{E}_t \left[\frac{C_{t+1}^{-1/\sigma}}{P_{t+1}^D} \right] \tag{26}$$

$$\frac{L_t^{1/\psi}}{C_t^{-1/\sigma}} = \frac{W_t}{P_t^D} = w_t. \tag{27}$$

The labour supply equation (27) implies that the real wage w_t equals the marginal rate of substitution between consumption and leisure when there is no friction in the labour market. The following subsection introduces real wage rigidity, however, which potentially generates a wedge between the real wage and the marginal rate of substitution.

2.5 Real Wage Rigidity

I introduce real wage rigidity in the labour market, which prevents the real wage from adjusting immediately to the labour market conditions. Mathematically, following Blanchard and Galí (2007), I assume that the actual real wage w_t is a weighted geometric mean of the lagged real wage w_{t-1} and the flexible real wage w_t^f:

$$w_t = w_{t-1}^{\psi_w} \cdot (w_t^f)^{1-\psi_w}, \tag{28}$$

where $\psi_w \in [0, 1]$ is the parameter that governs the degree of real wage rigidity with $\psi_w = 0$ corresponding to the perfectly flexible labour market with no real wage rigidity.

The real wage rigidity reflects the friction in the labour market, such as fixed contracts, employment regulations, and recruitment costs.[6]

2.6 Calvo and Myopic Price Setting

Both upstream and downstream firms adjust their individual nominal prices $P_t^U(j)$ and $P_t^D(i)$ according to the Calvo pricing model with probability θ^U and θ^D of no adjustment, respectively. The upstream industry firms determine their reset price P_t^{U*} by solving the standard expected discounted profit maximisation problem:

$$\max_{P_t^{U*}} \mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} \left[\left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{1-\varepsilon} - x_{t+\ell}^U \left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{-\varepsilon} \right] Q_{t+\ell} \tag{29}$$

where $q_t = Q_t P_{t+1}^D / P_t^D$, $q_{t+\ell|t} \equiv q_t \cdots q_{t+\ell-1}$ is the real discounted factor.[7] Take the FOC with respect to P_t^{U*}:

$$\mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} \left[(1-\varepsilon) \left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{-\varepsilon} + \varepsilon x_{t+\ell}^U \left(\frac{P_t^{U*}}{P_{t+\ell}^U} \right)^{-\varepsilon-1} \right] \frac{Q_{t+\ell}}{P_{t+\ell}^U} = 0, \tag{30}$$

which can be arranged as

$$P_t^{U*} = \frac{\varepsilon}{\varepsilon - 1} \cdot \frac{\mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} x_{t+\ell}^U (P_{t+\ell}^U)^\varepsilon Q_{t+\ell}}{\mathbb{E}_t \sum_{\ell=0}^{\infty} (\theta^U)^\ell q_{t+\ell|t} (P_{t+\ell}^U)^{\varepsilon-1} Q_{t+\ell}}. \tag{31}$$

In words, the optimal reset price for the intermediate good is a markup over a weighted average of the expected future real marginal cost, where the weights are determined by the Calvo parameter θ^U, the real discount factor $q_{t+\ell|t}$, and the future output $Q_{t+\ell}$. It follows

[6] Though out of scope of the paper, microfoundation can be provided by, for example, introducing search and matching mechanism with Nash bargaining wage negotiation as discussed by Hall (2005). MOVE THIS TO INTRODUCTION

[7] As a feature of Leontief production function, the marginal cost does not depend on the firm's own output, so the amount of produced intermediate goods $Q_{t+\ell}$ appears linearly in the objective function.

that the upstream industry price level P_t^U is a weighted average of the lagged price level P_{t-1}^U and the reset price P_t^{U*}:

$$P_t^U = [\theta^U P_{t-1}^U + (1-\theta^U) P_t^{U*}]^{\frac{1}{1-\varepsilon}}. \tag{32}$$

The downstream industry firms also adopt the same Calvo pricing model with a parameter θ^D for the probability of no price adjustment. Additionally, I allow deviations from rational expectations à la Gabaix (2020) and Minton and Wheaton (2023); the downstream firms are "myopic" in a sense that they discount the future expected marginal cost when setting their reset price, and hence put more weight on the current marginal cost than non-myopic firms do. To implement myopia, I simply replace the (rational) expectation operator $\mathbb{E}_t[\cdot]$ in the downstream firms' reset price setting problem with a *myopic expectation operator* $\tilde{\mathbb{E}}[\cdot]$ whose definition is given in the second subsequent subsection (since it is defined in terms of the log-deviation of variables from steady state):

$$\max_{P_t^{D*}} \tilde{\mathbb{E}}_t \sum_{\ell=0}^{\infty} (\theta^D)^\ell q_{t+\ell|t} \left[\left(\frac{P_t^{D*}}{P_{t+\ell}^D} \right)^{1-\varepsilon} - x_{t+\ell}^D \left(\frac{P_t^{D*}}{P_{t+\ell}^D} \right)^{-\varepsilon} \right] \frac{Y_{t+\ell}}{P_{t+\ell}^D}. \tag{33}$$

Except for the myopia, the FOC and the optimal reset price for the downstream industry are derived in the same way as the upstream industry.

2.7 Competitive Equilibrium

The formal definition of the competitive equilibrium of the model is follows.

Definition (Competitive Equilibrium)**.** *A competitive equilibrium of the model consists of prices* $\{P_t^D(i), P_t^U(j), W_t, i_t\}_{t=0}^\infty$ *and allocations* $\{C_t, Y_t(i), Q_t(j), M_t(j), L_t, N_t^D(i), N_t^U(j), Z_t, E_t, B_t\}_{t=0}^\infty$ *such that:*

1. *Households maximise utility with respect to consumption and bond holdings (i.e., the Euler equation* (26) *holds), while the real wage follows the partial adjustment process* (28)*, therefore employment is determined by firms' labour demand at the prevailing real wage* w_t*.*

2. *The upstream industry firms set prices optimally under Calvo with parameter* θ^U*, whereas the downstream industry firms set prices optimally under Calvo with parameter* θ^D *and myopic expectations.*

3. *The aggregate employment is* $L_t = N_t^D + N_t^U + E_t$*, determined by firms' labour demand at the wage* W_t*.*

4. *The intermediate goods market clears in aggregate* $Q_t = M_t$*, where* Q_t *is aggregate upstream output and* $M_t = \alpha^D \Delta_t^D Y_t$ *is aggregate demand for intermediate goods composite by the downstream industry.*

5. *The consumption goods market clears:* $Y_t = C_t$*.*

6. *The balance of payments equilibrium holds:* $E_t = s_t Z_t$.

7. *The bond market clears:* $B_t = 0$.

8. *The terms of trade* s_t *follows the exogenous AR(1) process* (1).

2.8 Log-linearisation

The model is log-linearised around the steady state as follows. Detailed derivation is provided in Appendix A.

First, I define the myopic expectation operator $\tilde{\mathbb{E}}_t[\cdot]$ for the downstream firms' price setting problem.

Definition (Myopic Expectation)**.** *Let* $\hat{v}_t$ *be the log-deviation of a variable* v_t *from its steady state value* $\bar{v}$, *i.e.,* $\hat{v}_t = \log(v_t/\bar{v})$. *The myopic expectation of* $\hat{v}_{t+k}$ *for* $k \geq 0$, *denoted by* $\tilde{\mathbb{E}}_t[\hat{v}_{t+k}]$, *is defined as*

$$\tilde{\mathbb{E}}_t[\hat{v}_{t+k}] = m_f^k \mathbb{E}_t[\hat{v}_{t+k}], \tag{34}$$

where $m_f \in [0, 1]$ *is the myopia parameter with* $m_f = 1$ *corresponding to the rational expectation case with no myopia, while* $m_f = 0$ *corresponding to the fully myopic case with no foresight at all.*

Second, the model requires a monetary policy rule to be closed. I use a standard log-linearised Taylor rule

$$\hat{i}_t = \phi \hat{i}_{t-1} + (1-\phi)(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t) \tag{35}$$

with parameters ϕ, χ_π, and χ_y.

Combined with the log-linearised Taylor rule and the exogenous AR(1) process for the terms of trade (1), the following equations complete the log-linearisation of the model:

$$\hat{y}_t = \hat{L}_t - \alpha^U \alpha^D \hat{s}_t \qquad \text{(aggregate production function)} \tag{36}$$

$$\hat{x}_t^D = \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1-\Phi_{DU})\hat{w}_t \qquad \text{(downstream real marginal cost)} \tag{37}$$

$$\hat{X}_t^D = \hat{x}_t^D + \hat{p}_t^D \qquad \text{(downstream nominal marginal cost)} \tag{38}$$

$$\hat{X}_t^U = \alpha^U \hat{s}_t + \hat{w}_t + \hat{p}_t^D \qquad \text{(upstream nominal marginal cost)} \tag{39}$$

$$\hat{w}_t^f = \frac{1}{\sigma}\hat{y}_t + \frac{1}{\psi}\hat{L}_t \qquad \text{(flexible real wage)} \tag{40}$$

$$\hat{w}_t = \psi_w \hat{w}_{t-1} + (1-\psi_w)\hat{w}_t^f \qquad \text{(real wage rigidity)} \tag{41}$$

$$\hat{\pi}_t = \hat{p}_t^D - \hat{p}_{t-1}^D \qquad \text{(CPI inflation)} \tag{42}$$

$$\hat{p}_t^D = \theta^D \hat{p}_{t-1}^D + (1-\theta^D)\hat{p}_t^{D*} \qquad \text{(downstream Calvo price level)} \tag{43}$$

$$\hat{p}_t^{D*} = \beta m_f \theta^D \mathbb{E}_t[\hat{p}_{t+1}^{D*}] + (1-\beta m_f \theta^D)\hat{X}_t^D \qquad \text{(downstream reset price)} \tag{44}$$

$$\hat{p}_t^U = \theta^U \hat{p}_{t-1}^U + (1-\theta^U)\hat{p}_t^{U*} \qquad \text{(upstream Calvo price level)} \tag{45}$$

$$\hat{p}_t^{U*} = \beta \theta^U \mathbb{E}_t[\hat{p}_{t+1}^{U*}] + (1-\beta\theta^U)\hat{X}_t^U \qquad \text{(upstream reset price)} \tag{46}$$

$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]). \qquad \text{(IS equation)} \tag{47}$$

MOVE TO APPENDIX IF WORD COUNT EXCEEDS Here, I derive the log-linearised equation for the downstream real marginal cost (37) to show the derivation of the internally calibrated parameter Φ_{DU}. Before log-linearisation, the downstream real marginal cost is given by

$$x_t^D = \left(1 - \alpha^D + \alpha^D \frac{P_t^U}{W_t}\right) w_t. \tag{11}$$

In the steady state, the nominal marginal cost of the upstream industry is $\bar{X}^U = \bar{W}$ from (5). Also, every firm in the upstream industry will set the optimal reset price, which is the steady state nominal marginal cost multiplied by the constant markup:

$$\bar{P}^U = \mu \bar{W} = \frac{\varepsilon}{\varepsilon - 1} \bar{W}, \tag{48}$$

where $\mu \equiv \varepsilon/(\varepsilon - 1)$ is the markup. It follows that the steady state real marginal cost of the downstream industry is

$$\bar{x}^D = (1 - \alpha^D + \alpha^D \mu)\bar{w}. \tag{49}$$

The log deviation of P_t^U/W_t is $\hat{P}_t^U - \hat{W}_t$, and since $w_t = W_t/P_t^D$, $\hat{W}_t = \hat{w}_t + \hat{p}_t^D$. Therefore,

$$\widehat{\left(\frac{P_t^U}{W_t}\right)} = \hat{p}_t^U - \hat{w}_t - \hat{p}_t^D. \tag{50}$$

Log-linearising x_t^D around the steady state gives

$$\begin{aligned}
\hat{x}_t^D &= \frac{\alpha^D \cdot \bar{P}^U/\bar{W}}{1 - \alpha^D + \alpha^D \cdot \bar{P}^U/\bar{W}} (\hat{p}_t^U - \hat{w}_t - \hat{p}_t^D) + \hat{w}_t & (51)\\
&= \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu} (\hat{p}_t^U - \hat{w}_t - \hat{p}_t^D) + \hat{w}_t \qquad \text{since } \mu = \bar{P}^U/\bar{W} & (52)\\
&= \Phi_{DU}(\hat{p}_t^U - \hat{w}_t - \hat{p}_t^D) + \hat{w}_t, & (53)\\
&= \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1 - \Phi_{DU})\hat{w}_t, & (54)
\end{aligned}$$

where

$$\Phi_{DU} = \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu}. \tag{55}$$

2.9 Calibration

I calibrate the model based on the UK economy around the 2022 Energy Crisis. The model period is monthly. Summary of the parameter values is presented in Table 1. Here, I describe the sources and rationales of the calibration choices for each parameter. I provide sensitivity analysis in Section 3.1 to show how the results change if different parameter values are chosen.

First, the three parameters in the households' utility function follow the standard in the

New Keynesian Literature: the discount factor β is set to 0.9984, implied by the annual real interest rate of 2%; elasticity of intertemporal substitution σ is set to 0.5; and the Frisch elasticity of labour supply ψ is set to 3.

In the production network, I set the elasticity of substitution ε to 11, which implies the standard New Keynesian steady state markup of $\mu = 1.1$. Meanwhile, I set the energy cost share in the upstream industry α^U to 0.375. The upstream industry in the model represents energy-intensive sectors that utilise imported energy as input to produce intermediate goods. Using the annual Supply and Use Tables from the ONS, I calculated the output-weighted average energy cost share across the eight most energy-intensive and intermediate-goods-producing sectors annually from 2017 to 2022.[8] I adopt the average of the annual figures for 2017-2019 as the steady state value for $\alpha^U = 0.375$. α^D indicates the intermediate input share in the downstream industry, which is set to be 0.4. This implies that the overall energy cost share in the final consumption goods is $\alpha^U \alpha^D = 0.15$. Φ_{DU} is derived as

$$\Phi_{DU} = \frac{\alpha^D \mu}{1 - \alpha^D + \alpha^D \mu} = 0.4231, \tag{56}$$

which is the cost-based share of the intermediate goods in the downstream industry's real marginal cost. The difference to α^D is that the upstream industry's markup on the intermediate goods is taken into account in Φ_{DU}.

Calvo pricing parameters are set to be $\theta^D = 11/12$ and $\theta^U = 3/4$, which imply the average price durations of 12 months and 4 months for the downstream and upstream industry, respectively. A strand of literature has estimated the price durations using micro-level price data and tested the validity of the Calvo pricing model. Using the US data, Nakamura and Steinsson (2008) find that the median duration of consumer price changes is 8-11 months excluding sales, while that of producer price changes for intermediate goods is 7.0 months, implying that prices for intermediate goods are more flexible than those for final consumption goods. ADD MORE? UK data: Bunn and Ellis (2012a), Bunn and Ellis (2012b), Zhou and Dixon (2019)

The myopia parameter for the downstream industry $m_f = 0.75$ implies that the myopic downstream firms behaviourally discount the future expected marginal cost when setting their reset price, such that only m_f^k of the k-period ahead expected marginal cost is taken into account. Gabaix (2020) adopts $m_f = 0.8$ in his quarterly model based on older estimates in the literature, which is equivalent to $m_f = 0.8^{1/3} \approx 0.928$ in the monthly model. However, given that his myopic parameter is applied to the aggregate industry, rather than only to the downstream firms in my model, the value of m_f in my model should possibly be lower than that in Gabaix (2020). Minton and Wheaton (2023) estimate the myopia parameter for firms vis-à-vis shocks in oil prices and non-oil commodity prices. They conclude that downstream

[8] Specifically, the eight sectors are Electric Power Generation (`CPA_D351`), Gas Manufacture & Distribution (`CPA_D352_3`), Industrial Gases, Inorganics, Fertilisers (`CPA_C20A`), Petrochemicals (`CPA_C20B`), Paper and Paper Products (`CPA_C17`), Glass, Ceramics, Stone (`CPA_C23OTHER`), Basic Iron and Steel (`CPA_C241_3`), and Other Basic Metals & Casting (`CPA_C244_5`). Detailed calculations are available in Appendix B.

firms are almost rational ($m_f \approx 1$) with respect to oil price shocks, while almost completely myopic ($m_f \approx 0$) in case of price changes in non-oil commodities, potentially due to the difficulty in formulating forward-looking expectations on far end of supply chains. In my model, upstream firms produce a variety of intermediate goods using imported energy, of which downstream firms may or may not be able to easily forecast future price changes. Therefore, for the main result, I adopt $m_f = 0.75$, which in the sensitivity analysis will be compared with the rational expectation case ($m_f = 1$) and the fully myopic case ($m_f = 0$).

Regarding the real wage rigidity parameter ψ_w, Blanchard and Galí (2007) consider two example values of 0.5 and 0.9 in their quarterly model. I adopt the monthly equivalent of the mean value of the two, $\psi_w = 0.7^{1/3} \approx 0.888$, which implies that the real wage adjusts by 11.2% of the gap between the flexible real wage and the actual real wage each month.

The monetary policy Taylor rule equation has three parameters, for which I set as $\phi = 0.9$, $\chi_\pi = 3$, and $\chi_y = 0.5$ following the literature standards under Taylor principle ($\chi_\pi > 1$).

Finally, ρ indicates the degree of persistence of the terms of trade $\hat{s}_t$, which follows an AR(1) process. At the time of writing, the natural gas prices after its peak in August 2022 exhibit significantly high persistence, which is problematic in discussing the greater persistence of inflation. Moreover, in monthly frequency, there is not enough observations of the natural gas prices post-crisis to empirically estimate its process. Consequently, I set a conservative value of $\rho = 0.8$, which implies that the size of an exogenous shock to the terms of trade decays half within 3.1 months.

3 Simulation Results and Discussion

As a quantitative exercise, I simulate the model using the calibrated parameter values and analyse the impulse response functions (IRFs) to an exogenous shock to the terms of trade. In period 0, the economy is hit by an exogenous shock that raises the terms of trade $\hat{s}_t$ by 100% from the steady state level, which is equivalent to doubling the energy price in domestic currency. Figures 2 and 3 display the model simulation results.[9]

First, Figure 2 compares the inflation persistence of the benchmark model and the paper's main model after an exogenous shock to the terms of trade, which itself follows an AR(1) process with a persistence parameter $\rho = 0.8$. In order to compare the decay rate and absolute size of the initial shock simultaneously, on this plot I normalise the inflation response of the paper's main model and the terms of trade shock in period 0 to one. The terms of trade $\hat{s}_t$ is shown in grey dot dashed line, from which one can see that half of the initial shock disappears within 3.1 months, and only 10% of the original shock remains after 10.3 months. Inflation response of the benchmark model is shown in black solid line, showing that the benchmark model fails to replicate the inflation persistence that the UK economy experienced during the 2022 Energy Crisis, as it goes back to the steady state faster than the terms of trade

[9] Although the model period is monthly, the impulse responses of CPI inflation and nominal rate are shown in the annualised values.

Table 1: Model Parameter Values

Parameter	Symbol	Value
Households		
Discount factor	β	0.9984
Elasticity of intertemporal substitution	σ	0.5
Frisch elasticity of labour supply	ψ	3
Production and cost structure		
Energy cost share (upstream)	α^U	0.375
Intermediate input share (downstream)	α^D	0.4
Elasticity of substitution	ε	11
Downstream's intermediate good-cost share	Φ_{DU}	0.4231 (implied)
Price setting		
Downstream Calvo probability	θ^D	11/12
Upstream Calvo probability	θ^U	3/4
Myopia parameter (downstream)	m_f	0.75
Labour market		
Real wage rigidity	ψ_w	$0.7^{1/3} \approx 0.888$
Monetary policy		
Interest-rate smoothing	ϕ	0.9
Inflation response	χ_π	3
Output response	χ_y	0.5
Shock process		
Terms of trade persistence	ρ	0.8

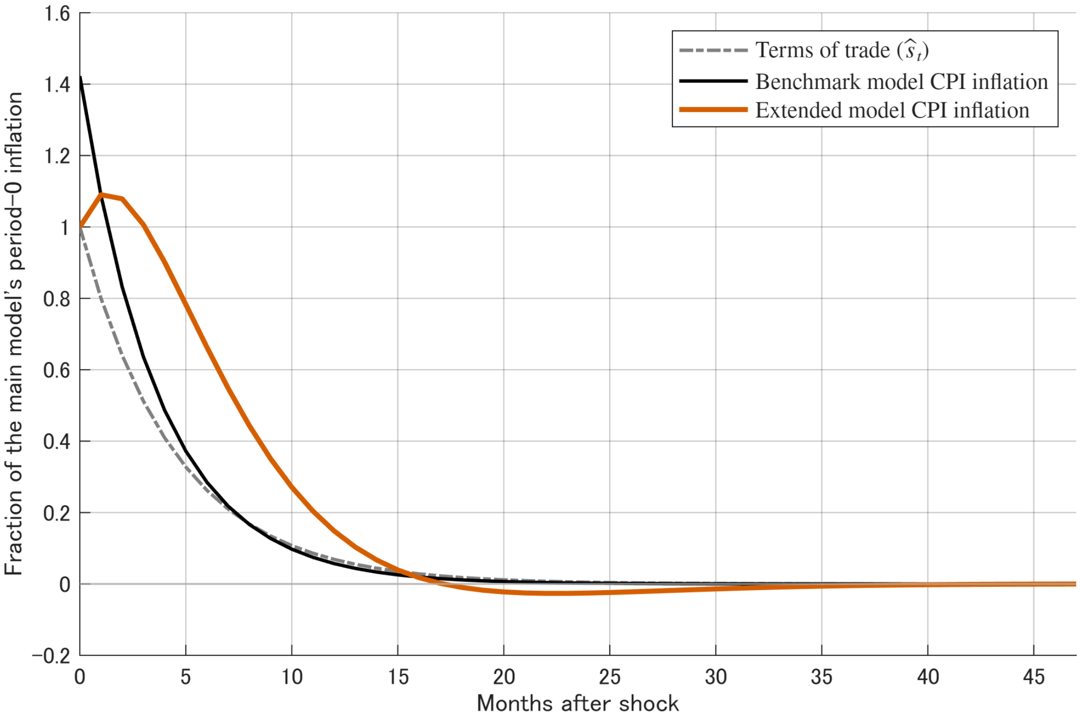

Figure 2: Inflation Persistence vs Energy Price Shock Decay

Notes: Impulse responses to a 100% terms-of-trade shock in period 0, which itself follows an AR(1) process with a persistence parameter $\rho = 0.8$ shown in grey dot dashed line. The inflation response of the benchmark model is shown in black, while that of the paper's main model is shown in orange. The inflation responses are normalised by the extended model's period-0 inflation.

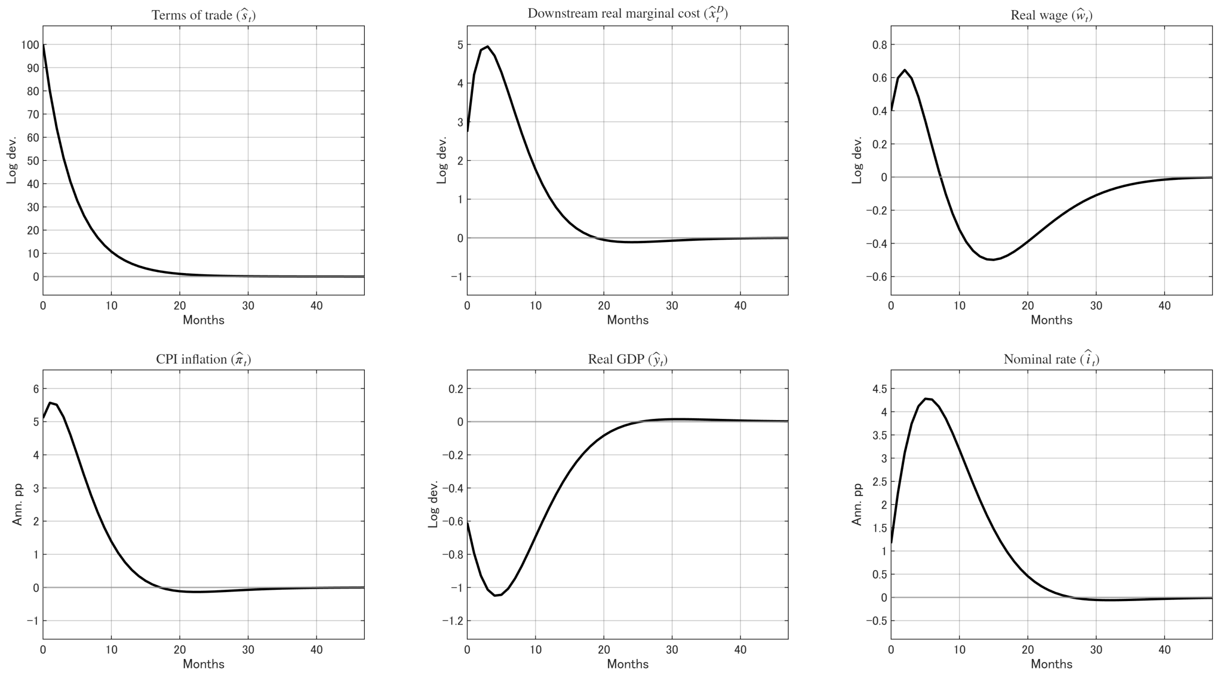

Figure 3: Impulse Responses of the Model to an Exogenous Terms of Trade Shock

Notes: Impulse responses to a 100% terms-of-trade shock in period 0 under the calibration in Table 1. CPI inflation and the nominal interest rate are reported in annualised percentage points; all other variables are in log deviations from steady state.

shock even though the initial shock to inflation is larger. In contrast, the paper's main model, whose inflation response is shown in orange solid line, exhibits a delayed peak of inflation one month after the initial shock, and the inflation rate decreases at a much slower pace than the benchmark model as well as the terms of trade shock with the half decay period of around 7.5 months.

Next, Figure 3 shows the IRFs of different endogenous variables in the main model, with which I describe the transmission mechanism of the energy price shock in generating the observed inflation persistence. First, on the top left panel, the terms of trade $\hat{s}_t$ is hit by a 100% exogenous shock in period 0, which decays monthly at the rate of $\rho = 0.8$, with only 10% of the initial shock remaining after 10 months.

As they use imported energy as input, the upstream industry firms experience an increase in their marginal cost, which leads to an immediate adjustment in the upstream price level $\hat{p}_t^U$ as the forward-looking Calvo-pricing firms increase their prices based on their rational expectations about the future path of the terms of trade. The increase in the upstream price level $\hat{p}_t^U$ then feeds into the downstream industry firms' real marginal cost $\hat{x}_t^D = \Phi_{DU}(\hat{p}_t^U - \hat{p}_t^D) + (1 - \Phi_{DU})\hat{w}_t$, shown on the top middle panel, through the cost-based input share of the intermediate goods $\Phi_{DU} \approx 0.42$. With a three-month lag, the downstream real marginal cost reaches the peak of 5.0% above the steady state level, after which it decreases gradually back to the steady state level.

Real wage $\hat{w}_t$ rises to 0.65% above the steady state level in the first few months. This is because the increase in the terms of trade $\hat{s}_t$ requires more labour to produce the tradable goods to maintain the balance of payments equilibrium. Even though the labour productivity essentially falls due to the increase in the energy price (as less labour can be allocated to the production of intermediate and consumption goods), which has a downward effect on the real wage, this effect is dominated by the upward pressure on the real wage from the increase in total employment; as more labour is needed to produce tradable goods, households move along their labour supply curve and require a higher real wage to compensate for the increased disutility of labour. Due to the real wage rigidity, the real wage adjusts slowly and reaches its peak with a two-month lag—which is why, as another factor of the downstream production, the downstream real marginal cost also peaks with a lag and decreases slowly afterwards. After the peak is reached, the real wage starts to decrease as the central bank raises the nominal interest rate $\hat{i}_t$ in response to the rising inflation $\hat{\pi}_t$, and demand falls subsequently. Importantly, the real wage rigidity brings the persistent effect of the terms of trade shock to both the upstream and downstream industry firms: for the upstream firms, even though the energy price decreases by 20% each month, the real wage adjusts more slowly, which delays the adjustment of the upstream marginal cost; for the downstream firms, the real wage rigidity lags the price adjustment of both inputs of the production directly and indirectly through the upstream price level, resulting in a persistent effect on the downstream marginal cost and inflation.

Myopia of the downstream firms also contributes to the inflation persistence. If downstream firms were rational, they would foresee that the positive deviation of the real marginal cost from the steady state level is only temporary, so that the optimal reset price would not be as high as in the myopic case, and the inflation would be less persistent. However, as the downstream firms are myopic with $m_f = 0.75$, they put more weight on the current real marginal cost than the rational expectation case, which makes inflation persistent even when the energy price shock itself decays relatively quickly.

On the bottom right panel, the nominal interest rate $\hat{i}_t$ increases by up to 4.2 percentage points in response to the rising inflation according to the Taylor rule. However, the monetary policy tightening is not sufficient to offset the inflationary pressure from the energy price shock, even with a cost of reducing the aggregate demand.

3.1 Sensitivity Analysis

I conduct sensitivity analysis with varying parameter values for examining the model robustness as well as for understanding the role of different mechanisms in generating the inflation persistence. Figure 4 shows the resulting persistence of inflation response under different parameter values to be compared with the original. Similarly to Figure 2, I normalise the main model's inflation response and the terms of trade shock in period 0 to one. On the top left panel, I vary the real wage rigidity parameter ψ_w from the original value of $0.7^{1/3} \approx 0.888$ shown in orange to 0 in blue and $0.9^{1/3} \approx 0.965$ in green. It shows that in absence of the real wage rigidity, the initial inflation response is higher than the original case with no delayed peak in the response, but the inflation decays monotonically at a faster pace than the original case. In contrast, with a higher degree of real wage rigidity of $\psi_w = 0.9^{1/3} \approx 0.965$, though the peak level of inflation is lower than the original case, the inflation response is more persistent. Therefore, comparing the three cases with different real wage rigidity parameters, one can see that the real wage rigidity plays a role in delaying the peak and generating persistence of the inflation response.

Second, the top right panel varies the downstream firms' myopia parameter m_f from 0.75 to 1 (completely rational, shown in blue) and 0 (fully myopic, shown in green). When the downstream firms are completely rational, the hump in the inflation response disappears and the inflation is less persistent than the original case. On the other hand, when the downstream firms are fully myopic, i.e, their reset price is solely based on the current marginal cost, the initial inflation response is smaller than the original case because of the lag in the upstream price adjustment, after which the inflation response increases and peaks with a three-month lag followed by a slower decay than the original case.

Next, on the bottom left panel, the Calvo probability parameter for the upstream industry θ^U is varied from the original value of $3/4$ to 0 and $11/12$ shown in blue and green, respectively. When the upstream price is completely flexible with $\theta^U = 0$, the initial inflation response is 2.7 times as large as the original case, which however decays at a much faster pace afterwards. This is because the upstream firms can adjust their prices instantaneously in response to the energy price shock, so that the shock is transmitted to the downstream firms immediately. In contrast, when the upstream price is more rigid with $\theta^U = 11/12$, only a small fraction of the upstream firms can adjust their prices in each period, so that the shock is transmitted to the downstream firms more slowly, which results in a smaller initial inflation response but a more persistent response with a longer lag in the peak. To conclude, the upstream Calvo probability parameter θ^U controls the tradeoff between the initial size of the inflation response and its persistence.

Finally, the bottom right panel adjusts the downstream firms' myopia parameter m_f and the real wage rigidity ψ_w simultaneously. Shown in pink is when the downstream firms are completely rational and there is no real wage rigidity, which informs that the vertical production network structure with Calvo pricing alone does not generate a hump-shaped response of inflation nor the inflation persistence. The blue line adds myopia of the downstream firms $m_f = 0.75$ but the labour market is still flexible, which enlarges the initial inflation response without exhibiting a hump-shaped response. Then, the original case with the real wage rigidity of $\psi_w = 0.7^{1/3} \approx 0.888$ is shown in orange, which displays that the real wage rigidity plays a role in delaying the peak of the inflation response and making the inflation more persistent, as observed in the UK economy after 2022.

4 Conclusion

Policy implications?

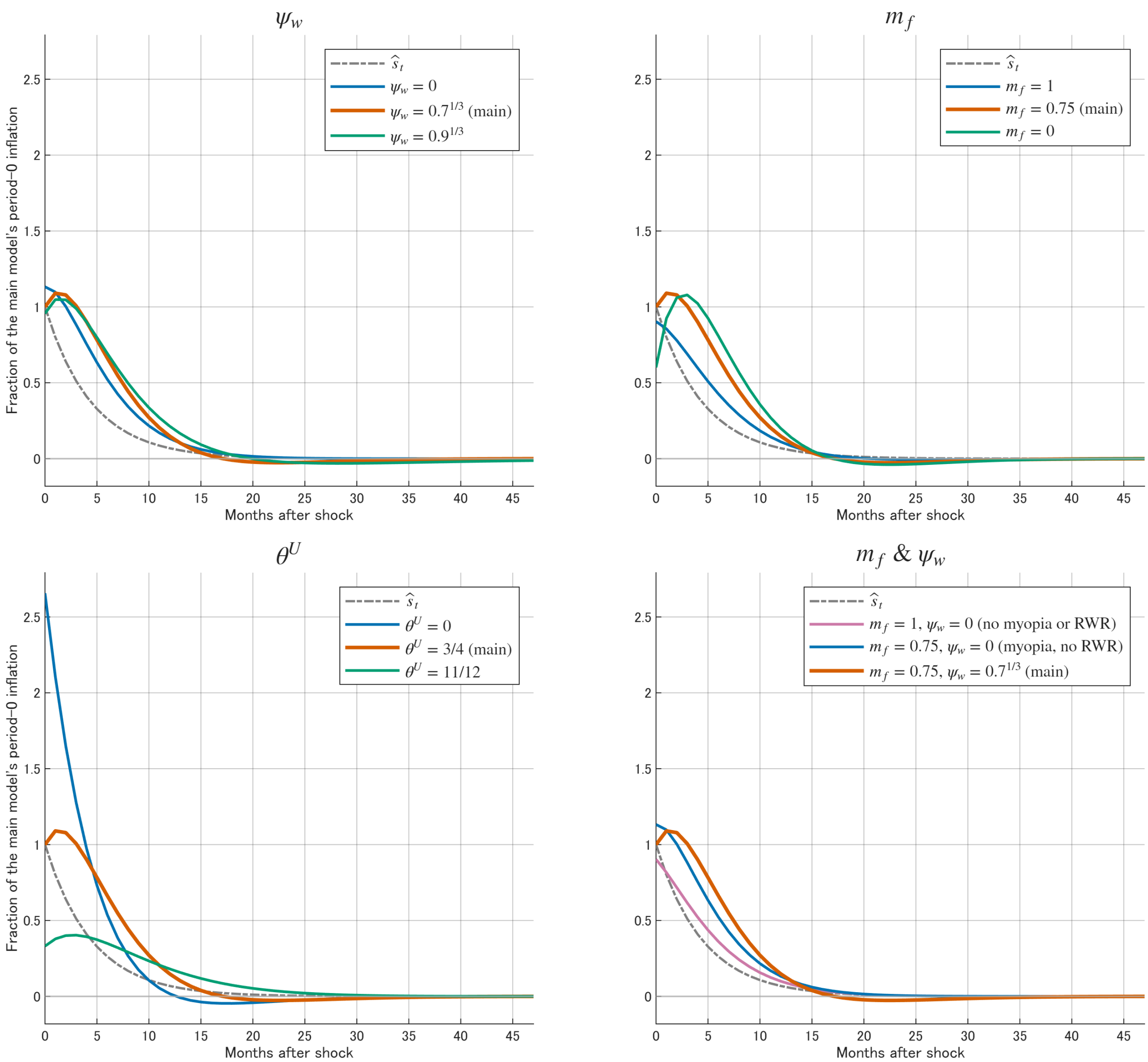

Figure 4: Sensitivity Analysis Results

Notes: Each panel shows the CPI inflation response to a 100% terms-of-trade shock under different parameter values. All series are normalised by the main calibration's period-0 inflation response. The terms of trade shock is shown in a grey dotted line.

https://www.ons.gov.uk/economy/nationalaccounts/supplyandusetables/datasets/ukinputoutputanalyticaltablesdetailed UK input-output tables, potentially useful to calibrate α^U and α^D.

Model limitations:

- Network structure of production and input-output linkages

Although the model assumes that the upstream industry firms imports energy by themselves, in reality, some firms in such industries may purchase energy from domestic energy retailers. However, in terms of price setting, it is still relevant since in data literature finds price of crude materials is almost completely flexible.

Model ignores changes in TFP either exogenous or endogenous by the energy price shock.

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A Appendix: Derivation of Log-linearised Equations

The log-linearised equations of the model are presented in (36)–(47). Among those equations, this appendix section derives the non-standard ones, namely the aggregate production function (36), the upstream nominal marginal cost (39), and the downstream reset price (44).

The aggregate production function (36). The aggregate output is

$$Y_t = \frac{L_t}{\left[1 - \alpha^D + (1 - \alpha^U + \alpha^U s_t)\alpha^D \Delta_t^U\right] \Delta_t^D}. \tag{22}$$

In steady state, $\bar{s} = 1$ and $\bar{\Delta}^U = \bar{\Delta}^D = 1$, so $\bar{Y} = \bar{L}$. The price dispersion terms Δ_t^D and Δ_t^U are second-order around the zero-inflation steady state, so $\hat{\Delta}_t^D = \hat{\Delta}_t^U = 0$ to a first-order approximation. Taking logs:

$$\log Y_t = \log L_t - \log\left[1 - \alpha^D + (1 - \alpha^U + \alpha^U s_t)\alpha^D\right].$$

Let $f(s_t) \equiv 1 - \alpha^D + (1 - \alpha^U + \alpha^U s_t)\alpha^D$. At $\bar{s} = 1$, $f(\bar{s}) = 1$ and $f'(\bar{s}) = \alpha^U \alpha^D$. Since $\bar{s} = 1$, $\hat{s}_t \approx s_t - 1$ to first order, so

$$\hat{y}_t = \hat{L}_t - \frac{f'(\bar{s})}{f(\bar{s})}\hat{s}_t = \hat{L}_t - \alpha^U \alpha^D \hat{s}_t. \tag{36}$$

The upstream nominal marginal cost (39). The upstream nominal marginal cost is

$$X_t^U = (1 - \alpha^U + \alpha^U s_t)W_t. \tag{5}$$

In steady state, $\bar{s} = 1$, so $\bar{X}^U = \bar{W}$. Taking logs and linearising around $\bar{s} = 1$:

$$\hat{X}_t^U = \frac{\alpha^U}{1 - \alpha^U + \alpha^U \bar{s}}\hat{s}_t + \hat{W}_t = \alpha^U \hat{s}_t + \hat{W}_t.$$

Since $w_t = W_t / P_t^D$, it follows that $\hat{W}_t = \hat{w}_t + \hat{p}_t^D$, so

$$\hat{X}_t^U = \alpha^U \hat{s}_t + \hat{w}_t + \hat{p}_t^D. \tag{39}$$

The downstream reset price (44). Following the same steps as for the upstream industry, the first-order condition of the downstream firm's reset price problem (33) yields

$$P_t^{D*} = \frac{\varepsilon}{\varepsilon - 1} \cdot \frac{\tilde{\mathbb{E}}_t \sum_{\ell=0}^{\infty} (\theta^D)^\ell q_{t+\ell|t}\, x_{t+\ell}^D (P_{t+\ell}^D)^\varepsilon Y_{t+\ell}}{\tilde{\mathbb{E}}_t \sum_{\ell=0}^{\infty} (\theta^D)^\ell q_{t+\ell|t} (P_{t+\ell}^D)^{\varepsilon-1} Y_{t+\ell}},$$

where $\tilde{\mathbb{E}}_t$ denotes the myopic expectation operator. Log-linearising around the zero-inflation steady state where $\bar{P}^{D*} = \mu \bar{X}^D$, the standard recursive representation of the Calvo reset price gives

$$\hat{p}_t^{D*} = \beta\theta^D \tilde{\mathbb{E}}_t[\hat{p}_{t+1}^{D*}] + (1 - \beta\theta^D)\hat{X}_t^D.$$

Note that the myopic expectation operator applies only to the forward-looking term $\hat{p}_{t+1}^{D*}$, not to the current nominal marginal cost $\hat{X}_t^D$ which is observed. Substituting the definition $\tilde{\mathbb{E}}_t[\hat{v}_{t+k}] = m_f^k \mathbb{E}_t[\hat{v}_{t+k}]$ with $k = 1$:

$$\hat{p}_t^{D*} = \beta m_f \theta^D \mathbb{E}_t[\hat{p}_{t+1}^{D*}] + (1 - \beta m_f \theta^D)\hat{X}_t^D.$$

To verify the coefficient on $\hat{X}_t^D$, iterate forward:

$$\begin{aligned}
\hat{p}_t^{D*} &= (1 - \beta m_f \theta^D) \sum_{\ell=0}^{\infty} (\beta m_f \theta^D)^\ell \mathbb{E}_t[\hat{X}_{t+\ell}^D] \\
&= (1 - \beta m_f \theta^D) \left[\hat{X}_t^D + \beta m_f \theta^D \mathbb{E}_t[\hat{X}_{t+1}^D] + (\beta m_f \theta^D)^2 \mathbb{E}_t[\hat{X}_{t+2}^D] + \cdots \right].
\end{aligned}$$

The weights sum to one, confirming that the reset price is a weighted average of current and expected future nominal marginal costs, where myopia ($m_f < 1$) increases the relative weight on current costs. When $m_f = 1$, this reduces to the standard rational-expectations Calvo reset price, which is the case for the upstream firms (46).

B Appendix: Calculations of Upstream Energy Cost Shares

For the main model calibration, I calculate the energy cost shares of the upstream industries to be $\alpha^U = 0.375$ based on the UK Input-Output Analytical Tables from the ONS. For the upstream industry candidates, the eight most energy-intensive sectors are selected, which are Electric Power Generation (`CPA_D351`), Gas Manufacture & Distribution (`CPA_D352_3`), Industrial Gases, Inorganics, Fertilisers (`CPA_C20A`), Petrochemicals (`CPA_C20B`), Paper and Paper Products (`CPA_C17`), Glass, Ceramics, Stone (`CPA_C23OTHER`), Basic Iron and Steel (`CPA_C241_3`), and Other Basic Metals & Casting (`CPA_C244_5`). Even though the Input-Output Analytical Tables reveal that those industries produce final consumption goods as well, the model considers them as upstream industries for simplicity.

Table A.1 reports the energy cost and gross output of the eight chosen industries between 2017 and 2019, as well as the resulting energy cost shares. At the bottom, average energy cost shares across the eight industries are reported, which are weighted by the gross output of each industry.

Table A.1: Energy Cost Shares of Upstream Industries between 2017 and 2019 (£m)

	2017			2018			2019		
Sector	Energy	Output	Share	Energy	Output	Share	Energy	Output	Share
Electric power generation	38,921	73,588	52.9	38,986	76,734	50.8	42,061	81,020	51.9
Gas manufacture & distribution	20,197	40,699	49.6	20,178	41,410	48.7	20,285	40,651	49.9
Industrial gases, inorganics, fertilisers	299	2,738	10.9	285	2,787	10.2	304	3,029	10.0
Petrochemicals	474	11,917	4.0	487	12,905	3.8	434	10,803	4.0
Paper and paper products	378	12,594	3.0	394	12,614	3.1	381	12,424	3.1
Glass, ceramics, stone	209	6,669	3.1	230	6,895	3.3	200	7,120	2.8
Basic iron and steel	215	7,825	2.7	244	7,932	3.1	208	8,242	2.5
Other basic metals & casting	140	6,272	2.2	203	6,586	3.1	141	6,015	2.3
Output-weighted average	60,832	162,302	37.5	61,007	167,863	36.3	64,014	169,304	37.8

Notes: Energy and Output columns report the sum of electricity (`CPA_D351`) and gas (`CPA_D352_3`) intermediate input costs and gross output, respectively, in millions of pounds. Share is energy cost as a percentage of gross output. Source: ONS Input-Output Analytical Tables.

Exact calculations with the identical eight industries are performed for the years between 2020 and 2022 as well, and the resulting energy cost shares are plotted in Figure A.1. It indicates that the energy cost share was somewhat stable until 2020, after which it increased significantly to 46.5% in 2022. I adopt $\alpha^U = 0.375$ which is within the value range between 2017 and 2020 and takes a conservative stance given the post-Covid upward trend for the model calibration.

C Appendix: The Benchmark Model

This appendix section presents the benchmark small open economy New Keynesian model, on which the paper's main model is extended.[10] Compared to the main model, the benchmark model does not have the vertical production network but entails a single industry that inputs imported energy and labour to produce final consumption goods according to their Leontief production function. The firms are completely rational and not myopic ($m_f = 1$), and there is no real wage rigidity ($\psi_w = 0$). The benchmark model is characterised by the

[10]The development of the benchmark model is due to Kevin Sheedy.

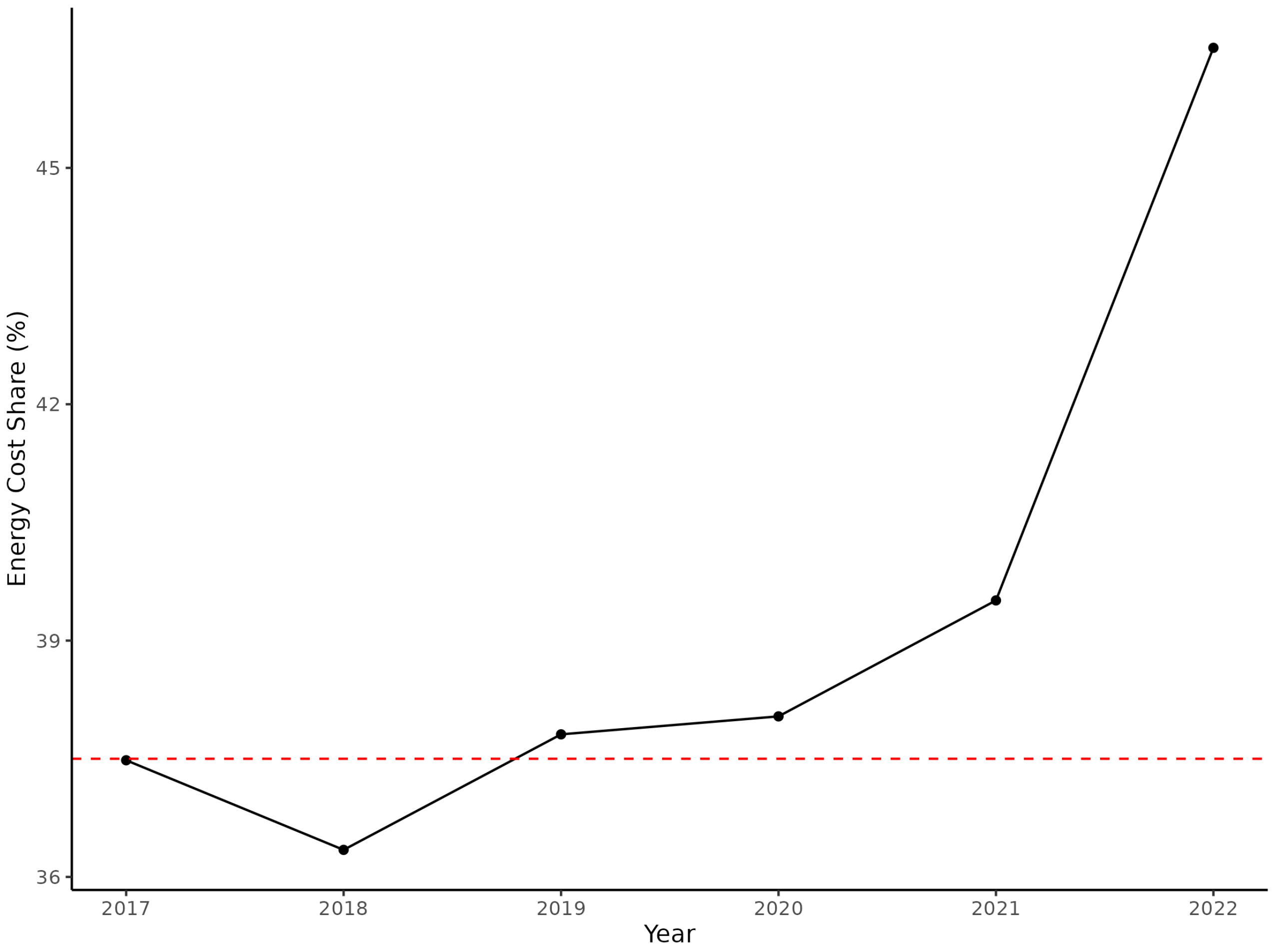

Figure A.1: Energy Cost Share of Upstream Industries

Notes: Energy cost share is calculated for each year between 2017 and 2022 and shown in black line. The red dashed line represents the adopted value $\alpha^U = 0.375$. Source: ONS Input-Output Analytical Tables.

following ten log-linearised equations:

$$\hat{y}_t = \hat{L}_t - \alpha \hat{s}_t \qquad \text{(aggregate production function)} \tag{A.1}$$
$$\hat{x}_t = \hat{w}_t + \alpha \hat{s}_t \qquad \text{(real marginal cost)} \tag{A.2}$$
$$\hat{X}_t = \hat{x}_t + \hat{P}_t \qquad \text{(nominal marginal cost)} \tag{A.3}$$
$$\hat{w}_t = \frac{1}{\sigma}\hat{y}_t + \frac{1}{\psi}\hat{L}_t \qquad \text{(labour supply)} \tag{A.4}$$
$$\hat{\pi}_t = \hat{P}_t - \hat{P}_{t-1} \qquad \text{(inflation)} \tag{A.5}$$
$$\hat{P}_t = \theta \hat{P}_{t-1} + (1-\theta)\hat{P}^*_t \qquad \text{(Calvo price level)} \tag{A.6}$$
$$\hat{P}^*_t = \beta\theta \mathbb{E}_t[\hat{P}^*_{t+1}] + (1-\beta\theta)\hat{X}_t \qquad \text{(reset price)} \tag{A.7}$$
$$\hat{y}_t = \mathbb{E}_t[\hat{y}_{t+1}] - \sigma\big(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]\big) \qquad \text{(IS equation)} \tag{A.8}$$
$$\hat{i}_t = \phi \hat{i}_{t-1} + (1-\phi)\big(\chi_\pi \hat{\pi}_t + \chi_y \hat{y}_t\big) \qquad \text{(Taylor rule)} \tag{A.9}$$
$$\hat{s}_t = \rho \hat{s}_{t-1} + u_t \qquad \text{(terms-of-trade shock)} \tag{A.10}$$

For calibration, the firms' energy cost share in production α is set to 0.15, which coincides with the overall energy cost share in the final consumption goods in the main model. The Calvo probability θ is set to 11/12, which I adopted for the downstream industry in the main model as well. Other parameters are set to the same values as in the main model as summarised in Table A.2.

Table A.2: Benchmark Model Parameter Values

Parameter	Symbol	Value
Households		
Discount factor	β	0.9984
Elasticity of intertemporal substitution	σ	0.5
Frisch elasticity of labour supply	ψ	3
Production and cost structure		
Energy cost share	α	0.15
Price setting		
Calvo probability	θ	11/12
Monetary policy		
Interest-rate smoothing	ϕ	0.9
Inflation response	χ_π	3
Output response	χ_y	0.5
Shock process		
Terms of trade persistence	ρ	0.8

The impulse responses of the benchmark model to an exogenous 100% shock to the terms of trade are shown in Figure A.2.

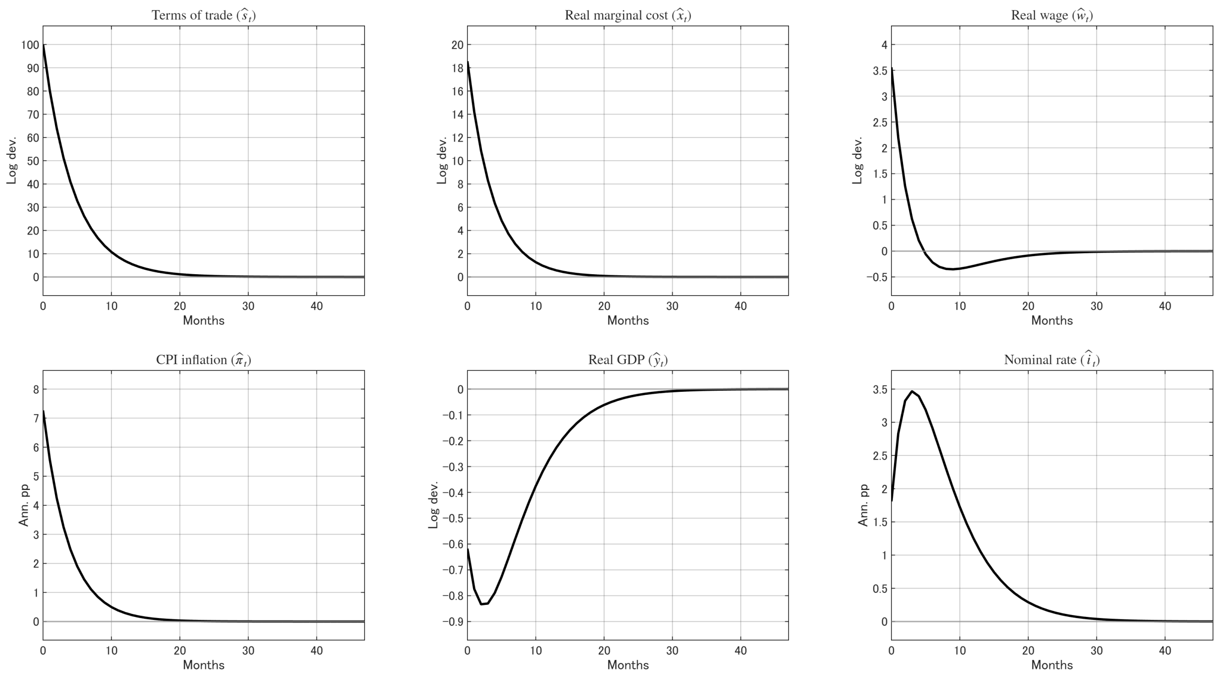

Figure A.2: Impulse Responses of the Benchmark Model to an Exogenous Terms of Trade Shock

Notes: Impulse responses to a 100% terms-of-trade shock in period 0 under the calibration in Table A.2. CPI inflation and the nominal interest rate are reported in annualised percentage points; all other variables are in log deviations from steady state.